

LESSON 7-3 • SESSION 1 • PRACTICE • FORM A

Absolute Value as Distance

Learning target – I can use absolute value to describe how far a number is from zero. • 6.NS.C.7c

REMEMBER

- **Absolute value** is how far a number is from 0. Write it with bars: $|-4| = 4$ and $|4| = 4$.
- Distance is never negative, so **absolute value is never negative**.
- **Greater** means farther right. **Farther from 0** means a bigger absolute value. -9 is farther from 0 than 4, but 4 is greater.

WORKED EXAMPLEFind $|-13|$ and $|6|$. Which number is farther from 0?**Step 1** -13 is 13 units left of 0, so $|-13| = 13$.**Step 2** 6 is 6 units right of 0, so $|6| = 6$.**Step 3** Compare the distances: $13 > 6$.**Answer** -13 is farther from 0**WORDS** **absolute value** (*valor absoluto*) • **distance** (*distancia*) • **magnitude** (*magnitud*)**PRACTICE**

show your work • label units

1. Find each absolute value.

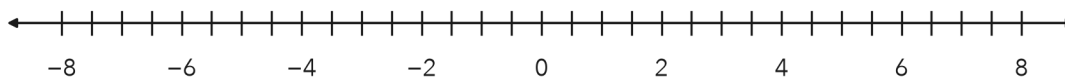
a. $|-16| = \underline{\hspace{2cm}}$

b. $|9| = \underline{\hspace{2cm}}$

c. $|0| = \underline{\hspace{2cm}}$

d. $|-2.75| = \underline{\hspace{2cm}}$

- 2.
- Beaker Readings.**
- Four beakers read: A
- -6
- degrees, B
- 2.5
- degrees, C
- -0.5
- degrees, D
- 8
- degrees.

Hint Count the units from 0 to each point.

a. Absolute values: A $\underline{\hspace{1cm}}$ B $\underline{\hspace{1cm}}$ C $\underline{\hspace{1cm}}$ D $\underline{\hspace{1cm}}$

b. Plot all four. Which beaker is closest to 0? Farthest? $\underline{\hspace{2cm}}$

- 3.
- Two Accounts.**
- One account has a balance of
- -45
- dollars. Another has a balance of
- 12
- dollars.

a. Absolute value of each balance: $\underline{\hspace{2cm}}$

b. Which account is closer to a zero balance? $\underline{\hspace{2cm}}$

Show your work.

- 4.
- Survey Cards.**
- One card reads
- -7.5
- m and another reads
- 7.5
- m. Find each absolute value. Do they mark the same place?

Hint Same distance does not mean same side.

Show your work.

Answer $\underline{\hspace{2cm}}$

- 5.
- Cold Storage.**
- A cooler logged
- -14
- degrees at midnight. Find the absolute value. What does it tell you?

Show your work.

Answer $\underline{\hspace{2cm}}$

- 6. Greater Magnitude.** Which number has the greater absolute value, -36 or 22 ? Justify your choice.

Hint Compare 36 and 22.

Show your work.

Number _____

- 7. Omar's Claim.** Omar says $|-5.2| = -5.2$ because the number inside the bars is negative. Is he right? Give the correct value.

Hint Can a distance be negative?

Show your work.

$|-5.2|$ _____

- 8. Bank Balances.** Three accounts read -85 , 40 , and -120 dollars.

- Find the absolute value of each balance. _____
- Which balance is farthest from 0? _____
- Which balance is greatest? Why is this a different question? _____

Show your work.

FIND THE ERROR

explain the mistake, then fix it

- 9. Bianca's Work.** Bianca said -19 is closer to 0 than 6: "The negative number is smaller, so it must be closer to zero." Explain her mistake. Then say which number is closer to 0.

TEST PRACTICE

choose the best answer

- 10.** $|-17|$ equals –

A -17 **B** 17 **C** 0 **D** $1/17$

- 11.** Which number is closest to 0?

A -9 **B** 7 **C** -3 **D** 5

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

-120 R	-36 U	22 Q	12 E	40 E	-85 Z	14 S	7.5 A	-14 X	16 M
#1a		#3b	#4	#5	#6	#8b	#8c		

CHECK YOURSELF

☐ I can do this on my own ☐ I need a little help ☐ I have a question about #_____